INSTALLATION GUIDE

Software Defect Prediction System

Step-by-step instructions to install and run the application on a new Windows computer. Written for complete beginners: no programming or command-line experience needed.

What you need	A Windows 10 or 11 computer (64-bit), internet, about 2 GB of free space, permission to install programs
Time	About 30-45 minutes the first time (mostly waiting for downloads); about 1 minute on later days
Result	The app runs on your computer and opens in your web browser at **http://127.0.0.1:5000**
Also covers	macOS and Linux (short appendix at the end)

The 4 parts of this guide

1. **Install the tools**: Python and Git (Steps 1-4)
2. **Get the project**: download it from GitHub (Steps 5-6)
3. **Set it up**: one-time preparation (Steps 7-11)
4. **Use it**: start the app, log in, analyse code (Steps 12-15)

Project: https://github.com/Debdeeptax/defect-predictor (private) · Guide version: 04 November 2026

Before you start: how to use this guide

You will mostly do two things: **install programs** (like any other program) and **type short commands** into a window called **PowerShell**. Every step tells you exactly what to type and what you should see afterwards. The coloured boxes mean:

> **Type this, then press Enter**
> `this is a command you type`

> **You should see**
> This is what the screen should show when the step worked.

> **If something goes wrong**
> - This tells you how to fix the most common problems for that step.

> **Tip**
> Extra advice that makes things easier.

Golden rules for typing commands

1. Type (or copy and paste) the command **exactly** as shown: same spelling, same spaces, same dots and backslashes (\).
2. Press the **Enter** key after each command. Nothing happens until you press Enter.
3. **Wait** until the command finishes. You know it is finished when a new line starting with `PS C:\...>` appears and the cursor blinks again.
4. Do one command at a time, in the order shown. Do not skip steps.
5. If a command shows red text, stop and read the *If something goes wrong* box for that step before continuing.

Copy and paste in PowerShell

Select the command in this PDF and copy it (`Ctrl+C`). Click inside the PowerShell window and paste with `Ctrl+V` or by **right-clicking**. Then press Enter.

Words you will see

Word	Meaning
PowerShell	A window where you type commands to the computer (it comes with Windows).
Command	One line of text you type into PowerShell and confirm with Enter.
Folder / path	A location on the disk, written like `C:\SDP\defect-predictor` .
Python	The programming language the app is written in. You install it once.
Git	A tool for downloading code projects. The app also uses it to analyse projects from the internet.
Virtual environment	A private folder (named .venv) that holds the app's own copy of the Python add-ons it needs, so it cannot conflict with other programs.
Server	The running app. It must stay running while you use the website.
127.0.0.1	Means "this computer". The address only works on the computer running the app.

Part 1: Install the tools

1 Step 1: Open PowerShell

~1 min

1. Click the **Start** button (Windows logo, bottom-left of the screen) or press the Windows key on the keyboard.
2. Type **PowerShell**.
3. Click **Windows PowerShell** in the results (a blue or black window opens).

You should see

A window with a line like `PS C:\Users\YourName>` and a blinking cursor. This is where you will type commands. You can leave it open for the whole guide.

Tip

Do **not** choose "Run as administrator". A normal PowerShell window is all you need. On some computers the app is called **Terminal** instead; that works too.

2 Step 2: Install Python 3.11

~5 min

The app needs **Python version 3.11** exactly (newer versions such as 3.12 or 3.13 have not been tested with it). Version 3.11.9 is the last 3.11 release with a normal Windows installer.

1. Open your web browser and go to:
   `https://www.python.org/downloads/release/python-3119/`
2. Scroll down to the **Files** table and click **Windows installer (64-bit)**. The file **python-3.11.9-amd64.exe** downloads.
3. Open the downloaded file (usually in your **Downloads** folder).
4. **Important:** at the bottom of the first screen, tick the box **"Add python.exe to PATH"**.
5. Click **Install Now**. If Windows asks "Do you want to allow this app to make changes?", click **Yes**.
6. When you see **Setup was successful**, click **Close**.
7. **Close PowerShell and open it again** (Step 1), so it notices the new program.

Check it worked: type this, then press Enter

```
py -3.11 --version
```

You should see something like

```
Python 3.11.9
```

If something goes wrong

- **'py' is not recognized**: Python is not installed correctly. Run the installer again, tick "Add python.exe to PATH", and make sure the option "py launcher" is selected. Then close and reopen PowerShell.
- **No suitable Python runtime found** or a different version number: you have another Python version but not 3.11. Install 3.11.9 as described above (other versions can stay).
- **"Windows protected your PC"** when opening the installer: click **More info**, then **Run anyway**. Only do this for files downloaded from python.org.

3 Step 3: Install Git

~3 min

1. Go to `https://git-scm.com/download/win` .
2. Click **Click here to download** (the 64-bit Git for Windows setup).
3. Open the downloaded file and click **Yes** if Windows asks for permission.
4. Click **Next** on every screen without changing anything, then **Install**, then **Finish**.
5. Close PowerShell and open it again.

Check it worked

```
git --version
```

You should see something like

```
git version 2.xx.x.windows.x
```

If something goes wrong

- **'git' is not recognized**: close every PowerShell window, open a new one and try again. If it still fails, run the Git installer again with the default options.

4 Step 4: (Optional) Install Google Chrome

~2 min

Any modern browser (Edge, Chrome, Firefox) can **use** the app. Google Chrome is only needed if you want to run the automatic browser tests (Selenium) later. If Chrome is already installed, skip this step. Otherwise download it from `https://www.google.com/chrome/` and install it with the default options.

Part 2: Get the project

5 Step 5: Get access to the private GitHub project

~5 min

The project is stored in a **private** GitHub repository, so only invited people can download it.

If you are the new user:

1. Create a free account at **https://github.com/signup** (if you do not have one).
2. Send your GitHub username to the project owner.
3. Open the invitation e-mail from GitHub (or go to https://github.com/notifications) and click **Accept invitation**.

If you are the project owner (Debdeeptax): open the repository on GitHub, click **Settings** → **Collaborators** → **Add people**, type the new user's username and click **Add**.

> **Tip**
> Alternative without a GitHub account: the owner downloads the ZIP file (Step 6, Option A) and gives it to you on a USB stick or by a file-sharing link. Then continue with Step 6, point 4.

6 Step 6: Download the project

~3 min

Choose **one** of the two options. Option A is the easiest.

Option A (recommended): download a ZIP file

1. Log in to GitHub and open `https://github.com/Debdeeptax/defect-predictor`.
2. Click the green **<> Code** button, then **Download ZIP**. A file named **defect-predictor-master.zip** downloads (about 50 MB).
3. Open your **Downloads** folder in File Explorer.
4. **Right-click** the ZIP file → **Extract All...**
5. In the box "Files will be extracted to this folder", delete what is there and type `C:\SDP`. Click **Extract**.
6. A window opens showing the folder **defect-predictor-master**. Right-click it → **Rename** → type `defect-predictor` → press Enter.

> **You should see**
> The folder `C:\SDP\defect-predictor` exists and, when you open it, contains files such as **run.py**, **requirements.txt**, **README.md** and folders **app**, **models**, **data**.

Option B: download with Git (for people who will update the project later)

Type these commands in PowerShell, one at a time:

> **Type these, pressing Enter after each line**

```
mkdir C:\SDP
cd C:\SDP
git clone https://github.com/Debdeeptax/defect-predictor.git defect-predictor
```

The first time, a window asks you to sign in to GitHub: click **Sign in with your browser** and approve. The download then finishes with a line like **Resolving deltas: 100%**.

If something goes wrong

- **Repository not found**: you have not accepted the invitation (Step 5) or you signed in with a different GitHub account.
- **A folder named defect-predictor already exists**: you already downloaded it. Use that folder, or delete it and try again.

Part 3: Set it up (only once)

All commands from now on are typed in PowerShell **inside the project folder**. Step 7 shows how to get there; you need to do Step 7 every time you open a new PowerShell window.

7 Step 7: Go into the project folder

~1 min

Type these, pressing Enter after each line

```
cd C:\SDP\defect-predictor
dir
```

You should see

The line now starts with `PS C:\SDP\defect-predictor>` , and `dir` lists names including **app**, **data**, **models**, **run.py** and **requirements.txt**.

If something goes wrong

- **Cannot find path**: the folder has a different name or location. Open File Explorer, find the folder that contains **run.py**, and check its name (it should be **defect-predictor** inside **C:\SDP**; see Step 6).
- You see only one folder called **defect-predictor-master** inside: the ZIP was extracted into an extra folder. Use cd once more to go into it, or move its contents up one level.

8 Step 8: Create the app's private Python folder (virtual environment)

~1 min

Type this, then press Enter

```
py -3.11 -m venv .venv
```

You should see

No message at all, just a new command line after a few seconds. A hidden-looking folder named **.venv** now exists in the project folder. That is correct.

If something goes wrong

- **No suitable Python runtime found**: Python 3.11 is missing; go back to Step 2.

9 Step 9: Install the add-ons the app needs

~5-15 min

This downloads about 250 MB of Python packages (machine learning, web server, PDF, and so on). Run the two commands one after the other:

Type these, pressing Enter after each line (wait for the first to finish)

```
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

You should see

Many lines starting with **Collecting ...** and **Downloading ...**, then a final line starting with **Successfully installed ...** followed by a long list of names. Yellow text saying a new pip version is available is harmless.

If something goes wrong

- **Read timed out**, **ConnectionError** or the download is extremely slow: your internet connection is slow or interrupted. Run the **retry command** below (it waits longer and tries again automatically). Parts that were already downloaded are reused.
- **The term '.\.venv\Scripts\python.exe' is not recognized**: you are not in the project folder (do Step 7) or Step 8 did not run.
- **Could not find a version that satisfies the requirement**: the virtual environment was created with the wrong Python. Delete the **.venv** folder in File Explorer and repeat Step 8 exactly as written.

Retry command for slow internet (only if needed)

```
.\.venv\Scripts\python.exe -m pip install --timeout 120 --retries 10 -r requirements.txt
```

10 Step 10: Create the settings file

~2 min

The app reads its settings from a file named **.env**. A ready-made example is included; you only copy it and set one secret value.

Type these, pressing Enter after each line

```
copy .env.example .env
.\.venv\Scripts\python.exe -c "import secrets; print(secrets.token_hex(32))"
```

You should see

The second command prints one long random line of letters and numbers (64 characters). This is your secret key. Select it with the mouse and copy it (Ctrl+C).

Open the settings file in Notepad

```
notepad .env
```

1. Notepad opens. Find the line **SECRET_KEY=change-me**.
2. Replace **change-me** with the long random line you copied, so it looks like **SECRET_KEY=3f9a1c...** (no spaces).
3. Save with **Ctrl+S** and close Notepad. Leave all other lines unchanged.

Tip

The secret key protects login sessions. Do not share it. Every computer can have its own key.

11 Step 11: Prepare the database (and create demo accounts)

~1 min

Type this, then press Enter

```
.\.venv\Scripts\python.exe scripts\reset_db.py
```

You should see something like

```
Database reset.
created developer dev@test.com / Test@1234
created qa        qa@test.com / Test@1234
created admin     admin@test.com / Test@1234
created sample report #1 for dev@test.com (8 files, 1 medium, 0 high)
```

Three demo accounts now exist, one for each role:

Email (username)	Password	Role	Can do
dev@test.com	Test@1234	Developer	Upload code, see own reports
qa@test.com	Test@1234	QA Engineer	See all reports, compare runs, export
admin@test.com	Test@1234	Admin	Everything, plus users, models, retraining, logs

If something goes wrong

- **No module named ...** (for example flask or sklearn): Step 9 did not finish. Run it again.
- Running this command again later is safe, but it **deletes all reports and accounts** and recreates the demo data. Use it to get a clean start.

Part 4: Use the app

12 Step 12: Start the app

~1 min

Type this, then press Enter

```
.\.venv\Scripts\python.exe run.py
```

You should see something like

```
  Software Defect Prediction System is running.
  Open http://127.0.0.1:5000 in your web browser.
  Keep this window open while you use the app. Press CTRL+C to stop it.

 * Serving Flask app 'app'
 * Debug mode: off
```

Leave this PowerShell window open. The app only works while it is running. You can minimise the window, but do not close it.

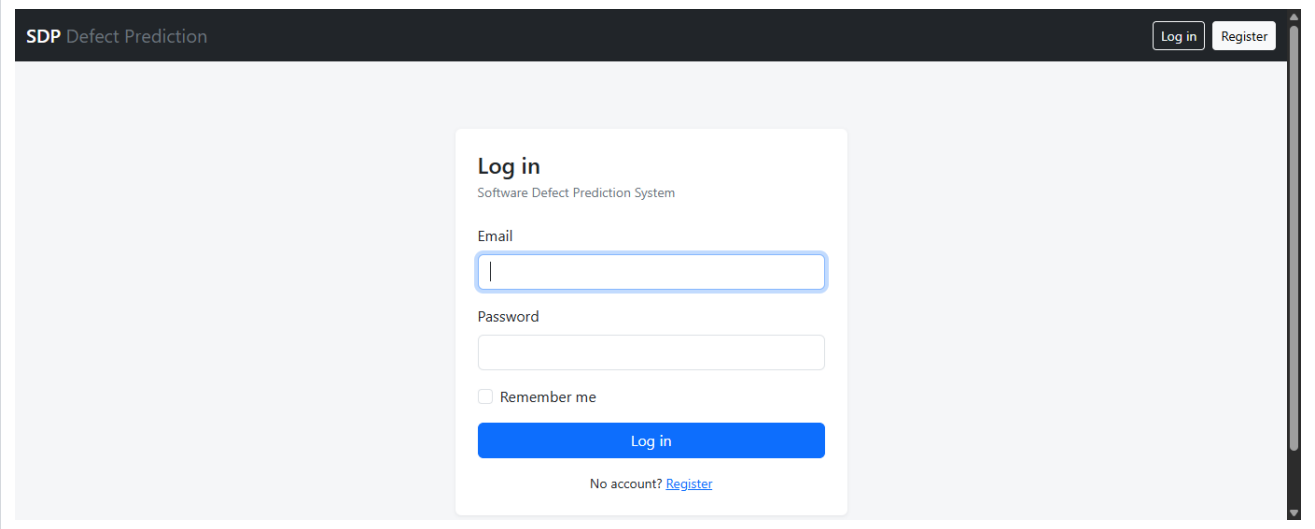
If something goes wrong

- A **Windows Security / Firewall** pop-up appears: you can click **Cancel**. The app is only used on this computer, so it does not need network access.
- **Address already in use** or **WinError 10048**: another program uses port 5000. Stop the app (Ctrl+C), type `notepad .env`, change **PORT=5000** to **PORT=5001**, save, start the app again and use **http://127.0.0.1:5001** in the browser.
- **No module named ...**: Step 9 is incomplete; run it again.

13 Step 13: Open the app in your browser and log in

~1 min

1. Open your web browser (Edge, Chrome, Firefox...).
2. Click the address bar at the top, type `http://127.0.0.1:5000` and press Enter.
3. The login page appears. Enter **dev@test.com** as email and **Test@1234** as password, then click **Log in**.



The login page.

You should see

A green message "Welcome back, Dev User!" and the **Dashboard** with counters and the active models.

If something goes wrong

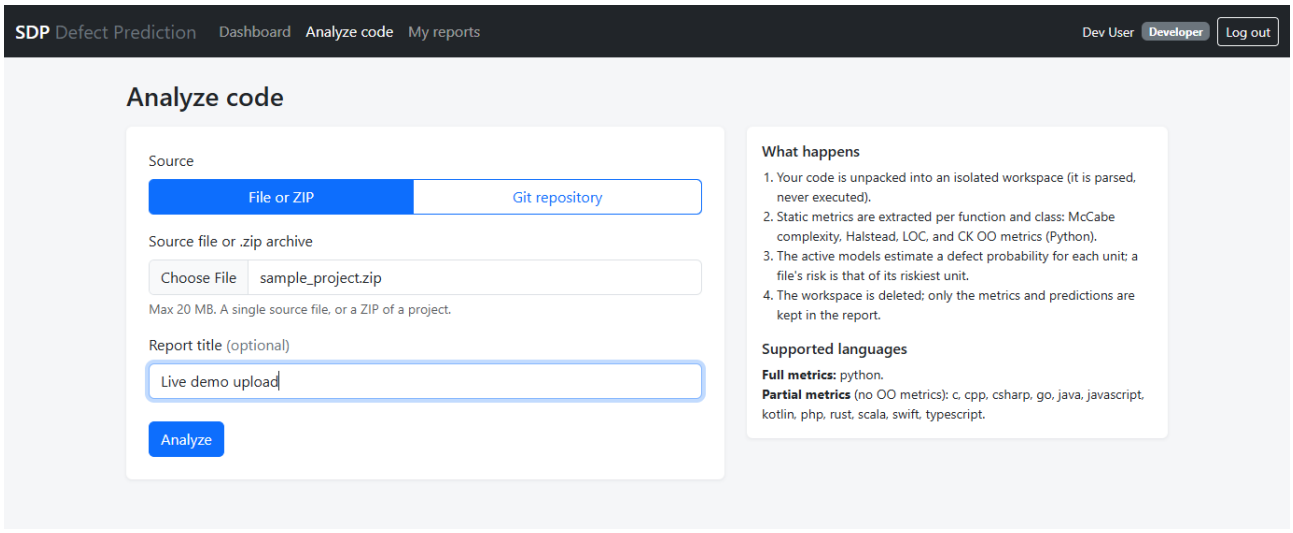
- **This site can't be reached**: the app is not running. Look at the PowerShell window from Step 12: if it was closed or shows an error, start the app again (Step 7, then Step 12).
- **Invalid email or password**: check the spelling (the password has a capital T and an @). If it still fails, stop the app, run Step 11 again, and start the app again.

14 Step 14: Analyse your first project

~2 min

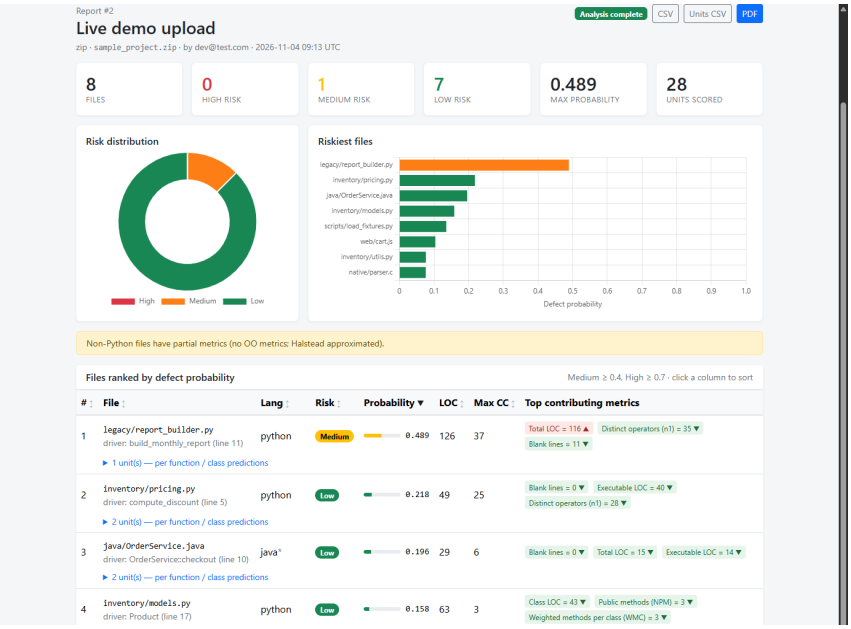
A small example project is included so you can try the app immediately.

1. Click **Analyze code** in the top menu.
2. Keep **File or ZIP** selected and click **Choose File**.
3. Go to `C:\SDP\defect-predictor\data\samples`, select **sample_project.zip** and click **Open**.
4. Optionally type a report title, then click the blue **Analyze** button.
5. Wait a few seconds while "Extracting metrics and predicting..." is shown.



The upload page with the sample ZIP selected.

> **You should see**
> A report page with a green **Analysis complete** badge, summary cards (8 files, 1 medium risk), two charts, and a table of files ranked by defect probability. **legacy/report_builder.py** is the riskiest file (Medium). Use the **CSV** and **PDF** buttons at the top right to download the report.



Top of a report page.

Tip
To analyse your own code: upload a single source file (for example a .py or .java file) or a ZIP of a whole project (up to 20 MB), or choose **Git repository** and paste the https address of a public GitHub, GitLab or Bitbucket project.

15 Step 15: Stop the app, and start it again another day ~1 min

To stop the app: click into its PowerShell window and press **Ctrl+C**. You can then close the window.

To start it again later (Steps 1-11 are never needed again):

1. Open PowerShell (Step 1).
2. Type the two commands below.
3. Open **http://127.0.0.1:5000** in your browser and log in.

Every time you want to use the app

```
cd C:\SDP\defect-predictor
.\.venv\Scripts\python.exe run.py
```

Tip
Your reports and accounts are kept between restarts. They are stored in the file instance\sdp.db inside the project folder.

Part 5: Checks, problems and extra information

16 Step 16: (Optional) Check that everything is healthy ~2 min

While the app is running, open `http://127.0.0.1:5000/health` in the browser.

You should see something like

```
{"database":true,"models":{"class":"class_v1_...joblib",
"function":"function_v1_...joblib"},"status":"ok"}
```

You can also run the automatic tests. Open a **second** PowerShell window (the app can keep running in the first one) and type:

Type these, pressing Enter after each line

```
cd C:\SDP\defect-predictor
.\.venv\Scripts\python.exe -m pytest -m "not selenium"
```

You should see
After about one minute, a line ending in **passed** in green (for example "160 passed"). If Google Chrome is installed you can run all tests, including the browser tests, with `.\.venv\Scripts\python.exe -m pytest`.

If something goes wrong

- If **health** shows **"status":"degraded"**, the model files are missing. The folder **models** must contain two .joblib files of about 16 MB and 27 MB. Download the project again (Step 6).

Common problems and solutions

What you see	What it means	What to do
'py' is not recognized	Python is not installed or not found	Step 2 (tick "Add python.exe to PATH"), then reopen PowerShell
No suitable Python runtime found	Python 3.11 is not installed	Install Python 3.11.9 (Step 2)
'git' is not recognized	Git is not installed	Step 3, then reopen PowerShell
Cannot find path 'C:\SDP\defect-predictor'	The folder has another name or place	Check the folder in File Explorer (Steps 6-7)
'.\.venv\Scripts\python.exe' is not recognized	Not in the project folder, or Step 8 skipped	Do Step 7, then Step 8
Read timed out / very slow download	Slow internet	Repeat Step 9 with --timeout 120 --retries 10
No module named 'flask' (or other name)	Add-ons not installed	Repeat Step 9
This site can't be reached	The app is not running	Start it (Steps 7 and 12); keep the window open
Address already in use / WinError 10048	Port 5000 is busy	Set PORT=5001 in .env (Step 12 box)
Invalid email or password	Typo, or the database was changed	Check spelling; or run Step 11 again
Prediction model unavailable / degraded	Model files missing	Download the project again (Step 6)
Red error text you do not understand	Something else failed	Take a screenshot of the whole window and send it to the project owner

Keeping it safe

- The demo password **Test@1234** is published in this guide. If other people can reach your computer, or you put the app on a server, replace the demo accounts: create your own accounts with strong passwords on the **Register** page, log in as admin, give them roles on the **Users** page, and **Deactivate** the demo accounts there.
- The app never runs the code you upload; it only reads it.
- The address 127.0.0.1 is only reachable from your own computer.
- Do not share your **.env** file; it contains your secret key.

Appendix: macOS and Linux (short version)

The steps are the same; only the commands differ. Install Python 3.11 (macOS: installer from python.org; Ubuntu/Debian: `sudo apt install python3.11 python3.11-venv git`) and Git, download the project as in Step 6, then open the **Terminal** app and run:

> **macOS / Linux commands (one per line)**
>
> ```
> cd ~/SDP/defect-predictor
> python3.11 -m venv .venv
> .venv/bin/python -m pip install --upgrade pip
> .venv/bin/python -m pip install -r requirements.txt
> cp .env.example .env          # then edit SECRET_KEY in .env
> .venv/bin/python scripts/reset_db.py
> .venv/bin/python run.py
> ```

Quick reference card (Windows)

All commands in order. Type each line in PowerShell and press Enter.

When	Command
Check tools (once)	`py -3.11 --version` `git --version`
Go to the project (every time)	`cd C:\SDP\defect-predictor`
Create virtual environment (once)	`py -3.11 -m venv .venv`
Install add-ons (once)	`.\.venv\Scripts\python.exe -m pip install --upgrade pip` `.\.venv\Scripts\python.exe -m pip install -r requirements.txt`
Settings file (once)	`copy .env.example .env` `notepad .env` (set SECRET_KEY)
Database + demo accounts (once, or to reset everything)	`.\.venv\Scripts\python.exe scripts\reset_db.py`
Start the app (every time)	`.\.venv\Scripts\python.exe run.py`
Open the app	`http://127.0.0.1:5000`
Stop the app	Press Ctrl+C in the app's PowerShell window
Health check	`http://127.0.0.1:5000/health`
Run tests (optional)	`.\.venv\Scripts\python.exe -m pytest -m "not selenium"`

Login	Password	Role
dev@test.com	Test@1234	Developer
qa@test.com	Test@1234	QA Engineer
admin@test.com	Test@1234	Admin

More details for developers: the README.md file in the project folder, and the project page https://github.com/Debdeeptax/defect-predictor.